

University of Essex

MSc Artificial Intelligence

Module: IA_PCOM7E January 2026

Unit 3: Collaborative Discussion – Agent-Based Systems

Initial Discussion

The discussion at hand revolves around two topics, each addressed in turn:

1. What are the main factors that led to the development of agent-based systems?
2. What benefits can such systems provide to enterprises and other organizations?

For the first topic, I would assert that agent-based systems have received increasing attention as a response to the need to develop solutions to problems that are, by their nature, unamenable to static or monolithic solutions (Goonatilleke and Hettige, 2022). In other words, when an unchanging, “one size fits all” solution cannot succeed due to the distributed nature of an issue, or the complexity of the issue when encountered at scale, or due to rapidly-changing environmental circumstances, agentic solutions have been developed to fill the gap (Goonatilleke and Hettige, 2022).

For the second topic, agentic solutions offer a variety of advantages. By their nature, they are designed to operate and interact with each other in complex environments with minimal human intervention (Pulikottil et al, 2023). This provides significant opportunities for finding solutions more quickly, at lower cost, and in a more resilient or future-proof manner, as an efficiently-structured agentic system will be intentionally designed to maximize its resilience, and to self-correct, when faced with complex factors in its environment that a static or monolithic solution might not be able to recognize in the first place, or – even if detected – respond to appropriately or effectively (Pulikottil et al, 2023; Tshakwanda, Arzo and Devetsikiotis, 2023).

References

Goonatilleke, S. and Hettige, B. (2022). Past, Present and Future Trends in Multi-Agent System Technology. *Journal Européen des Systèmes Automatisés*, 55(6), pp.723–739. doi:<https://doi.org/10.18280/jesa.550604>.

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Peer Reponse No. 1

I appreciate how your post is clear and straight to the point, framing the topic at hand in two concise questions.

For the first topic – as you framed it – I would add internal technological factors of related technologies advancement (Priyadarshi, 2025), whether on the side of infrastructure, like the overall increase computational capabilities, or the advent of natural language processing (NLP) and large language models (LLMs). I would argue that interaction with the machine through natural language due to the advancement in latter areas made agents and agent-based systems (ABS) more accessible and, therefore, more adoptable, and gave them one their main traits that Wooldridge (2009) asserted in the first place, which them being 'socially able'. Furthermore, I would add that ABS has special glamour around its capability of scaling, connecting and communicating with other agents and systems. This area still yields great promises as it is still a wide area for live research and continuous development (Joshi, 2025).

For the second topic, I agree with you and emphasise that the future of ABS, after all, would still be hinged on whether they would be able to increase profits and lower costs, through various ways, including rapid problem-detection and decision making, reducing the 'bottleneck' for the need of human intervention like the traditional software architecture, as well as reducing the need for wide labour with diversified set of skills (Joshi, 2025). However, I would prefer to be suspicious on how would business frame the limits between reducing the need for human intervention, and how much delegation of decision making to machines would actually prove to be useful and sensible.

As much it is exciting to witness and discuss the rise and proliferation of ABS, as much it raises a lot of suspicions and second thoughts, probably like the rise of any new technology.

References

Joshi, S. (2025). Review of Autonomous and Collaborative Agentic AI and Multi-Agent Systems for Enterprise Applications. *International Journal of Innovative Research in Engineering and Management (IJIREM)*, 12(3), pp. 65–76. Available at: <https://doi.org/10.55524/ijirem.2025.12.3.9>.

Priyadarshi, M. (2025). Autonomous AI Agents Transforming Enterprise Operations: From Static Automation to Intelligent Decision-Making Systems. *Sarcouncil Journal of Multidisciplinary*, 5(7), pp. 863–870. Available at: <https://doi.org/10.5281/zenodo.16408261>.

Wooldridge, M. (2009). Intelligent Agents. *An Introduction to MultiAgent Systems – Second Edition*. West Sussex, United Kingdom: John Wiley & Sons, pp. 21–48.

Discussion Summary

My initial post addressed two topics of discussion:

1. The primary factors that drove the development of agent-based systems;
2. The benefits that agent-based systems may provide to enterprises.

For the first topic, I asserted that the rise of agent-based systems is based in large part on enterprise needs to manage distributed issues at scale (Goonatilleke and Hettige, 2022).

For the second topic, I asserted that these systems provide more flexibility, customization, and lower development costs than conventional monolithic solutions (Pulikottil et al, 2023; Tshakwanda, Arzo and Devetsikiotis, 2023). I also asserted that efficiently-designed and built automated agents should be able to operate in their subject matter domains with minimal human oversight or intervention, which could help limit the risk of human error (Pulikottil et al, 2023).

Peer discussion raised additional salient points; first, since agent-based systems are often built from natural language processing foundations, they can be more “socially able,” or amenable to managing issues that revolve around human-software interaction, especially where those same interactions are premised on human users using conversational language that the agents must be capable of parsing and processing in an actionable manner to achieve their purposes (Joshi, 2025; Priyadarshi, 2025; Woolridge, 2009).

Secondly, peer discussion addressed the potential limitations of agent-based systems; namely, the risks inherent in delegating work conventionally performed by human actors to an automated system that, by its nature, lacks the ability to make any assessments or take any action outside of the scope of what it was designed to do (Joshi, 2025; Priyadarshi, 2025). This limitation could easily result in a scenario where anticipated efficiencies at scale, or expected cost savings from reduced human labor requirements, do not materialize; in a worst case scenario, the need for human oversight over a complex, distributed, automated system may even erase any initial productivity gains it may provide (Joshi, 2025; Priyadarshi, 2025).

References

Goonatilleke, S. and Hettige, B. (2022). Past, Present and Future Trends in Multi-Agent System Technology. *Journal Européen des Systèmes Automatisés*, 55(6), pp.723–739. doi:<https://doi.org/10.18280/jesa.550604>.

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